

Global Cybersecurity Threats (2015-2024)

DATA VISUALIZATION TECHNIQUES

Exercise 1

Group 7

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Company and Objective

- Company Name
Cyber Insecurity Consulting
- Industry
Cyber security
- Functional Area
Risk Assessment and Management
- Objective and Scope
Helping clients identify cyber security risks and develop mitigation strategies by researching cyber security threats and incidents.

The Dataset

Dataset URL:

<https://www.kaggle.com/datasets/atharvasoundankar/global-cybersecurity-threats-2015-2024>

1. Definition

Main Dataset:

The **Global Cybersecurity Threats Dataset (2015–2024)** is a comprehensive collection of recorded cyber incidents over a 10-year period, covering detailed information about attack types, methods, targeted industries, countries affected, and associated damages. It serves as a foundational resource for understanding the evolving landscape of cyber threats worldwide.

The dataset captures key attributes such as:

- **Threat Type** (e.g., Malware, DDoS)
- **Attack Vector** (e.g., Phishing, SQL Injection)
- **Affected Industry** (e.g., Finance, Healthcare)

- **Severity Level** (Low, Medium, High, Critical)
- **Financial Impact** (estimated loss in millions)
- **Response Time** (hours to mitigate)
- **Mitigation Strategy** (countermeasures applied)

Secondary Dataset:

The **Country Reference Table (2015–2024)** provides supplementary contextual information to support the analysis of global cybersecurity threats. It contains **approximate** or **projected** data over 10 years for 10 major countries, covering the following key indicators:

- **GDP (in Trillion USD)**
Estimated based on publicly available data from the World Bank for certain years, extended with reasonable projections to cover the full range from 2015 to 2024.
- **Internet Penetration (%)**
Based on historical data from the International Telecommunication Union (ITU) and the World Bank, with linear projections to estimate trends through 2024.
- **Cybersecurity Ranking**
Simulated data reflecting each country's relative cybersecurity posture. Since ITU's Global Cybersecurity Index (GCI) does not provide complete year-by-year rankings, the figures are approximated for illustrative purposes.

This information allows organizations, researchers, and policymakers to assess risks, predict trends, and devise effective cybersecurity strategies.